Kritansh Uppal

Visual design, branding & media

CREATIVE & DESIGN RESUME

Computer Science & Engineering Student at Punjab Engineering College · Chandigarh, India

GitHub: github.com/kirat11X LinkedIn: linkedin.com/in/kritansh-uppal

Email: kritansh.11.sss@gmail.com LeetCode: leetcode.com/u/kirat11X

CodeChef: codechef.com/users/kirat11X

Codeforces: codeforces.com/profile/kirat11X

PROFILE

Creative Head at WOPI, where I lead branding and visual strategy for an international academic conference.

I work across photography, photo and video editing, motion graphics, UI/UX design in Figma and graphic design in Canva — alongside a Computer Science & Engineering degree at Punjab Engineering College, which lets me build the tools behind the visuals too.

LEADERSHIP

Creative Head, WOPI

Led branding and visual strategy for an international academic conference.

Assistant Cultural Clubs Secretary (ACCS), Punjab Engineering College (PEC)

Liaise between student cultural clubs and college administration, coordinating official cultural initiatives.

Joint-Head, PECFEST

Coordinated logistics for 10+ events and managed 30+ volunteers serving 10000+ attendees at the college's flagship festival.

SKILLS

Visual & design	Photography, Photo editing, Video editing, Motion graphics, UI/UX design in Figma, Graphic design in Canva

Languages	Python, C++, Go, JavaScript, SQL

EXPERIENCE

Research Intern, TBRL, DRDO

May 2026 – Jul 2026 · Chandigarh

Built automated blast-fragment kinematic analysis pipelines using high-speed imagery, enabling fragment detection, multi-object tracking, and velocity estimation.

- Built a CV pipeline (YOLO, ByteTrack, optical flow, geometric methods) for fragment detection, tracking, and velocity estimation from high-speed imagery — achieving 70% detection with velocity accuracy within ±10% after calibration.
- Built and deployed a 5-stage end-to-end pipeline behind a Streamlit dashboard, containerized with Docker for local server testing, and packaged as a standalone air-gapped Windows installer (70 dependencies bundled) requiring zero internet connectivity.
- Researched impact-energy estimation from target-surface damage by integrating crater segmentation, depth reconstruction, and physics-based energy modeling of fragment impacts.

Python · Computer Vision · YOLO · ByteTrack · OpenCV · Streamlit · Docker · RTMPose

Software Intern, Zimo One

Jun 2025 – Jul 2025 · Remote

Engineered backend services in Node.js and implemented REST API validation suites for internal tools.

- Resolved 15+ production bugs in Node.js backend services, improving internal tool uptime by 12% for workflows used by 30+ team members.
- Implemented REST API validation using schema-based checks, reducing regression issues by 25% across multiple services.

Node.js · REST APIs · JavaScript · Git · Schema Validation

EDUCATION

Punjab Engineering College, Chandigarh

Aug 2024 – May 2028 (Expected)

Bachelor of Technology (B.Tech), Computer Science & Engineering
CGPA: 8.82 / 10

- Core coursework: Data Structures & Algorithms, Computer Vision, Machine Learning, Operating Systems, Database Management.

SELECTED PROJECTS

Automatic Short-Form Video Generator (Vizard-AI Replica)

Prototype · 2025

An automated computer vision clipping pipeline that extracts adaptive keyframes from long videos, cutting editing time by 70% and processing 5+ hours of video per run.

Python · faster-whisper · librosa · OpenCV · MediaPipe · sentence-transformers · FFmpeg · yt-dlp

Source on GitHub (github.com/kirat11X/video--clipper-vizard-AI-replica-automatic-short-form-clip-generator-)

ACHIEVEMENTS

- Winner, PECathon 2025 — Open-to-all inter-college hackathon conducted during PECFEST with 50+ teams.
- CodeChef Rating: 1568 | Codeforces Rating: 1329 | LeetCode Contest Rating: 1612.
- Awarded by Punjab Governor Sh. Gulab Chand Kataria ji for excellence in sports.
- State Gold Medalist — Table Tennis, Chandigarh.

INTERESTS

Computer Vision & Deep Learning, Generative AI & LLMs, Autonomous Video Pipelines, Data Structures & Algorithms, Full-Stack System Architecture, Multimodal Generative AI, Computer Hardware (GPUs & CPUs), Table Tennis, Photography & Cinematography, UI/UX Design